

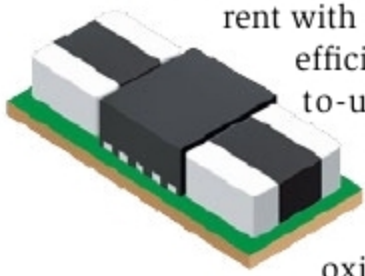
NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

DC/DC step-down power module

Texas Instruments has introduced a 5.5V step-down power module that delivers true, continuous 6A output current with up to 95 per cent efficiency. The easy-to-use TPSM82480 DC/DC module integrates power metal-oxide semiconductor



field-effect transistors (MOSFETs) and shielded inductors into a tiny, low-profile footprint for space- and height-constrained applications such as point-of-load telecommunications, networking and test-and-measurement power supplies.

TPSM82480 maintains the required 6A output current over the full temperature range without additional airflow. This is accomplished using a two-phase control topology that shares the load between the phases to ensure high efficiency and balanced operation. Additional features include adjustable soft start, voltage selection to support multiple processor stages and a power indicator.

Texas Instruments
www.ti.com

Single-channel gate driver ICs

Infineon Technologies offers EiceDRIVER 1EDC compact 300 mil family of single-channel gate driver integrated circuits (ICs). With an insulation test voltage of $V_{ISO} = 2500 \text{ V(rms)}$ for 60 seconds, these galvanically-isolated drivers are recognised under UL 1577.



Due to their high switching frequency of up to 1000kHz, these do not only drive IGBTs, but can be integrated in SiC MOSFET

topologies. A wide variety of applications profit from these characteristics, including photovoltaic string inverters, uninterruptible power supplies, electric vehicle charging stations, industrial drives, welding equipment, and commercial, construction and agricultural vehicles.

Infineon Technologies India Pvt Ltd
www.infineon.com

Amplifiers

With MAX98357 and MAX98358 digital-input class D audio power amplifiers from Maxim Integrated, designers can achieve highest efficiency in a compact, cost-efficient, plug-and-play solution. Despite their ultra-small form factors, these class D efficient amplifiers deliver 3.2W class AB audio performance for superior audio quality.

Maxim Integrated Products
www.maximintegrated.com

Hall-effect sensor IC

Allegro MicroSystems has introduced a high-precision, back-biased, programmable Hall-effect sensor IC. Allegro's ATS344 includes differential Hall elements, and is packaged in a custom



back-biased package that includes an integrated magnet. This device incorporates a 2-wire current-mode pulse-width modulation (PWM) output to help minimise pin count for remote sensors. It also includes on-chip EEPROM technology, capable of supporting up to 100 read/write cycles, for flexible end-of-line programming of calibration parameters. ATS344 is ideal for automotive applications requiring high-resolution

detection of long-stroke ($> 5\text{mm}$) linear motion.

Allegro MicroSystems Inc.
www.allegromicro.com

Programmable power management ICs

Renesas Electronics has announced three programmable power management ICs (PMICs) that offer high power efficiency and small footprint for application processors used in smartphones and tablets. These are ISL91302B, ISL91301A and ISL91301B PMICs.



The PMICs deliver power to artificial intelligence (AI) processors, field-programmable gate arrays (FPGAs) and industrial microprocessors. These are ideal for powering supply rails in solid-state drives, optical transceivers, and a wide range of consumer, industrial and networking devices. ISL91302B dual/single output, multi-phase PMIC provides up to 20A of output current and 94 per cent peak efficiency in a 70mm² solution size. This is more than 40 per cent smaller than competitive PMICs.

Renesas Electronics Corp.
www.renesas.com

IHLP inductors

Vishay Intertechnology has introduced a series of low-profile, high-current IHLP inductors that meet the requirements of MIL-STD-981 class S. Designed to provide high reliability in space-grade applications, inductors in Vishay Custom Magnetics SGIHLP series feature a ruggedised package and perform at high operating temperatures up to $+180^{\circ}\text{C}$.



Featuring a magnetically-shielded construction, the devices are optimised for DC/DC converters and high-current